

Matthew Xie

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Canadian citizen, TN-eligible

CORE COMPETENCIES

Founding ML engineer with proven track record of building and scaling end-to-end ML infrastructure in high-velocity startups and engineering complex Generative AI applications for production environments.

- **Generative AI & Retrieval Systems:** Architected complex LLM and Computer Vision workflows, semantic retrieval pipelines, and context-aware applications using frameworks like PyTorch, LangGraph and OpenCV.
- **ML Data Infrastructure:** Directed end-to-end product delivery, from data lakehouse ingesting 5.4M frames/week across production IoT fleet to training and distributing ML models to hardware.
- **Data Quality & Monitoring:** Built automated weekly diagnostics across all production sources, covering distribution drift detection, label mismatch analysis, and root-cause classification separating upstream data faults from hardware and model faults.
- **Core Stack:** Highly proficient in Python, C++, PyTorch, RKNN, OpenCV, and modern relational/NoSQL databases.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

Nov 2025 - Present

Uma

Toronto, ON

- Operated as **sole ML architect** in a high-velocity startup environment, singlehandedly designing and provisioning the **AWS data ecosystem** from the ground up while training production ML models to drive **rapid, end-to-end product delivery**.
- Architected **data lakehouse on AWS using Terraform, Apache Iceberg, Athena, and Kinesis Firehose** to ingest and query per-frame multimodal telemetry across a 19-machine fleet; schema captures per-frame embeddings, detection telemetry, tracker states, and camera diagnostics at **5.4M production frames/week**.
- Built **entire production real-time ML execution stack and tracking pipeline** for automated retail, spanning detection model, training pipeline, edge inference, and auto-labeling with **7 hour train-to-production cycle**.
- Spearheaded scaling ML infrastructure, designing **self-correcting MLOps pipeline** performing automated weekly diagnostics via **embedding drift detection, confidence distribution monitoring, and label mismatch analysis** across IoT fleet.
- Developed **zero-shot auto-labeling pipeline** combining MediaPipe hand tracking, YOLO-world, SAM2.1, and Kalman filtering to automatically create training datasets, **eliminating the need for manual data annotation** at 95%+ bounding box accuracy.
- Wrote C++ edge inference runtime to run models on Rockchip NPU and optimized throughput by partitioning camera inputs across multiple NPU cores, achieving 20ms latency per frame for 9.5 million parameter CV model.

Machine Learning Engineer Intern

May 2025 - Oct 2025

Themis AI

Boston, MA (remote)

- Led the design and engineering of **end-to-end active learning training pipelines** in PyTorch that achieved 90% of baseline accuracy using 5% of original training data through uncertainty-based sampling.
- **Improved upon prior published methods** by enhancing feature-space sampling by replacing PCA+clustering with ResNet-18 embeddings, achieving 2.5x greater accuracy gain over baseline random sampling.
- Implemented custom training loops, data loaders, and retraining cycles distributed over multiple GPUs.

Machine Vision Engineer Intern

Jan 2025 – Apr 2025

Zebra Technologies

Montreal, QC

- Engineered and deployed high-performance C++ applications for real-time streaming data processing, OCR, and edge execution on IoT hardware.
- Rewrote C++ compute codebase** to use scheduling and parallel operations, gaining **6x speedup** to real-time image processing.
- Implemented 2D+3D computer vision algorithms such as convolutional edge finding and point cloud merging.

Undergraduate Research Assistant

Jun 2024 – May 2025

University of Toronto

Toronto, ON

- Researched prompting and agentic methods for LLMs to solve **dynamic reasoning problems** by formulating Markov Decision Processes (MDP) and generating code solutions.
- Designed **chain-of-thought pipeline using LangChain** to improve MDP formulation accuracy by 60%.
- Constructed **RAG pipelines using LlamaIndex** to extract 5000 datapoints from PDF textbooks; contributed to evaluation and prompt engineering for agentic pipeline across GPT-4o, DeepSeek-R1, and Llama 3.3-70B.

Machine Learning Research Intern

May 2024 – Sep 2024

Pardee Lab

Toronto, ON

- Built **PyTorch training infrastructure** supporting 15+ transformer and LLM model architectures to predict various protein properties.
- Implemented **LoRA/QLoRA quantization** to reduce model size by 90% while maintaining 90% accuracy, allowing 15B-parameter model to be trained on a single local lab GPU for protein research.
- Optimized data preprocessing pipeline, achieving 12x speedup through parallel batching.

Software Engineer Intern

Jan 2023 - Apr 2023

Fundserv Inc

Toronto, ON

- Developed Angular website in TypeScript for client access to Azure backend through microservice APIs.
- Created novel accessibility testing tool for Angular and integrated it with existing CI/CD pipeline.
- Expanded Jasmine and Karma unit tests, increasing coverage from 20% to 95%.

PROJECTS

Loregraph – Efficient Graph Memory Engine

- Architected Loregraph, a local-first bi-temporal knowledge graph engine that retains superseded facts with validity windows, enabling **complex natural language point-in-time queries** over how knowledge changed.
- Replaced per-hop LLM graph traversal with personalized PageRank diffusion over a temporally-filtered subgraph, cutting **per-query cost to \$0.003 and reducing token cost of context by 20x at 700ms latency**.
- Achieved **86% on LongMemEval-S** and ships as an MCP server with embedding and extraction running fully local on-device using a tiny text classification model.

AI Customer Call Service

- Developed a real-time AI voice assistant using Twilio, and GPT-4o mini API for phone-based inquiries.
- Deployed a serverless architecture with AWS Lambda and DynamoDB for low-latency, scalable performance.
- Automated CI/CD pipeline with GitHub Actions to build Docker image, push to AWS ECR, and update Lambda.

SKILLS

Languages: Python, C, C++

Libraries & Frameworks: PyTorch, AWS, Terraform, LangGraph, NumPy, Pandas, scikit-learn, OpenCV, RKNN

Machine Learning: Transformers, RAG, LLM Agents, CNNs, Object Detection

Research: 3 co-authored papers in ML workshops (ICLR 2025, NeurIPS 2025) and conference (SmallSat 2024)

EDUCATION

University of Toronto

Honours Bachelor of Science in Statistics

Toronto, ON

Jun 2026